Fire Incident Trend Analysis

Singapore Fire Statistics & SafeEdge Detection Context

Prepared for: Firefighters & Fire Safety Officers

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SafeEdge

Edge-Based Fire Safety Intelligence System

DLW 2026 | Track 3: AI in Security | NTU Singapore

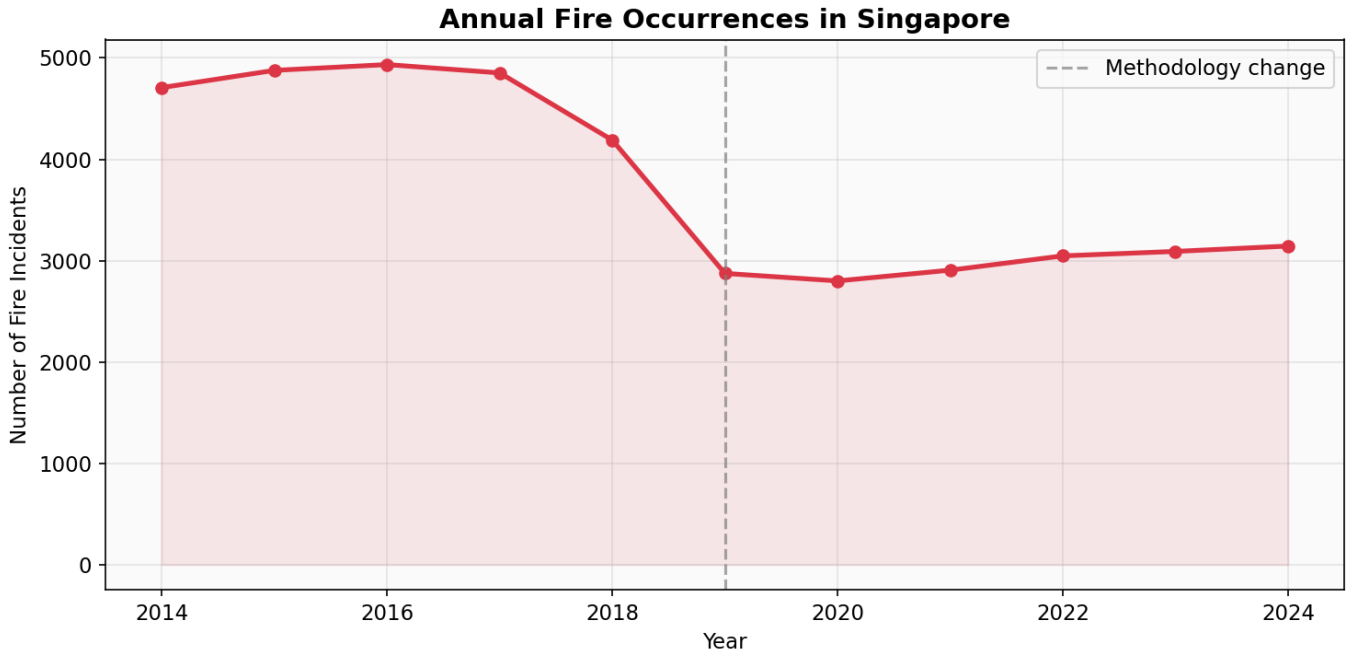
Executive Summary

In 2024, Singapore experienced a total of 3,144 fire incidents, reflecting a 1.7% increase from the previous year. The predominant category of these incidents was residential fires, which accounted for approximately 1,160 cases, underscoring the ongoing challenges in fire safety within domestic environments. Notably, the year saw 53 reported fire-related injuries and a single fatality, highlighting the critical need for targeted prevention strategies and public awareness campaigns to mitigate risks associated with residential fires.

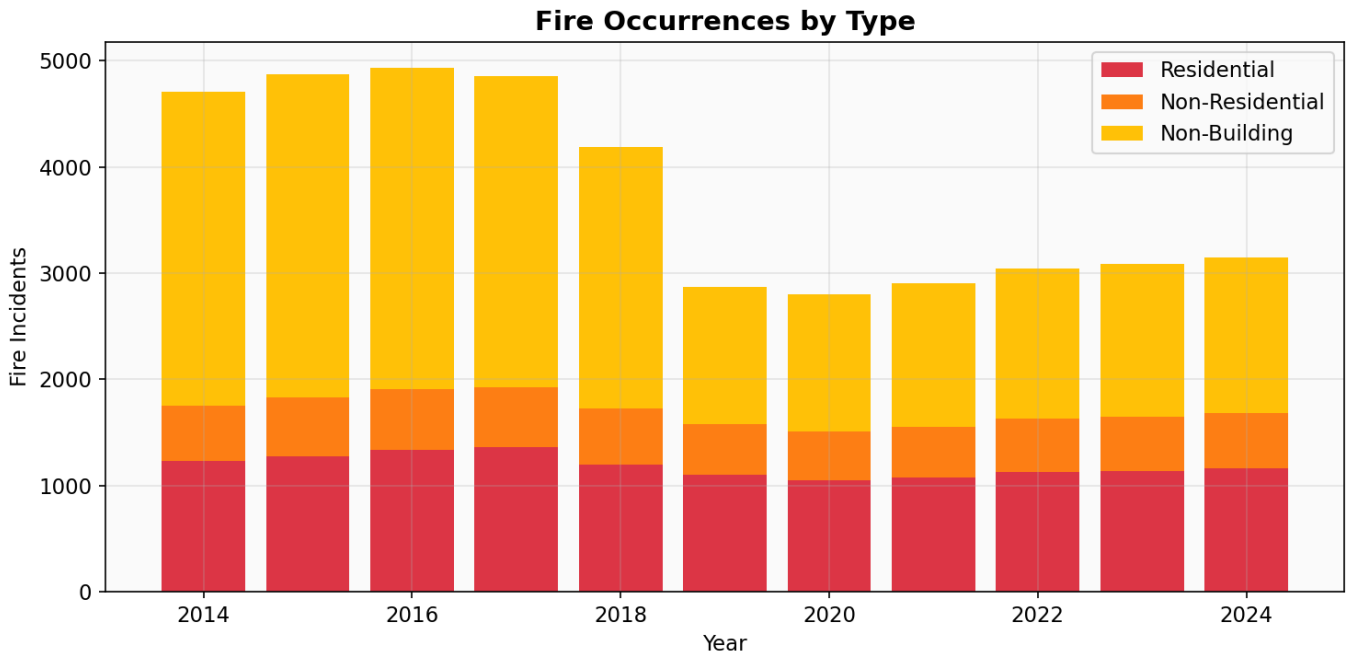
3,144	2024	+1.7%	1
Total Fires	Year	YoY Change	Fatalities

Annual Fire Occurrence Trends

Fire incidents have exhibited significant variability from 2014 to 2024, with reported occurrences ranging from 2,800 to 4,934. A notable decline in incidents was observed around 2019, attributed to changes in data collection methodologies that likely enhanced reporting accuracy. However, following this period, there has been a gradual increase in fire incidents post-2020, indicating a potential resurgence in fire activity that warrants further investigation and proactive response measures.



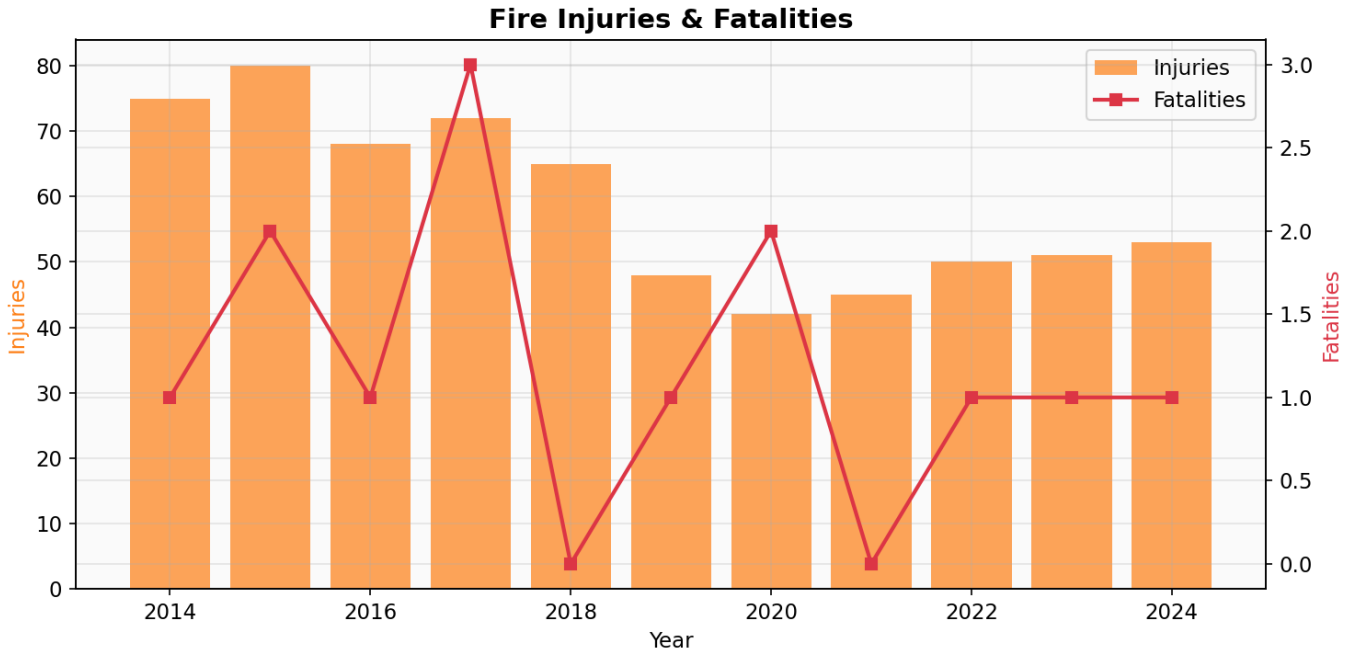
Annual fire occurrences in Singapore (Source: SCDF via Data.gov.sg)



Fire incidents by category: residential, non-residential, non-building

Fire Severity Analysis

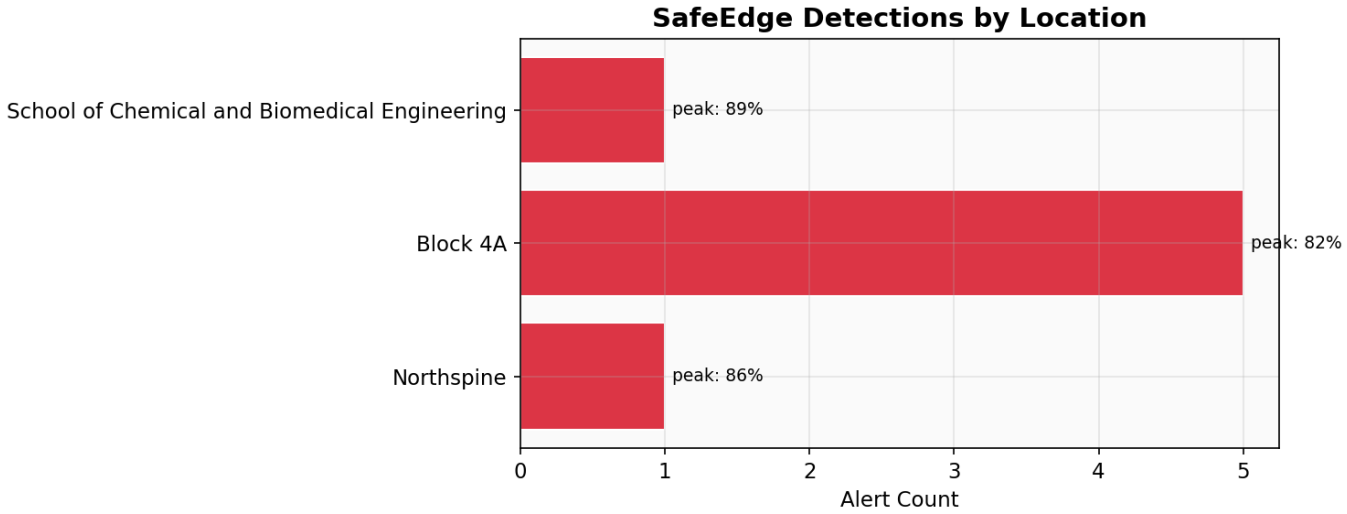
The analysis of fire-related injuries over the past three years indicates a concerning upward trend, with injuries increasing from 50 to 51 and then to 53. Conversely, fatalities have remained stable at one per year during the same period. The implementation of early detection systems, such as SafeEdge, has the potential to significantly reduce response times, thereby enhancing the likelihood of preventing injuries and fatalities in future incidents.



Fire injuries (bars) and fatalities (line) by year

SafeEdge Detection Context

During the reporting period, SafeEdge technology successfully detected a total of seven events across various locations on the NTU campus, specifically within the School of Chemical and Biomedical Engineering, Block 4A, and Northspine. The system exhibited a peak confidence level of 88.9%, indicating a high degree of reliability in its detection capabilities. Furthermore, the average detection frame rate was recorded at 8.2 frames per second, suggesting that the system maintained a consistent monitoring performance throughout the observation period. These metrics underscore the effectiveness of SafeEdge in enhancing campus safety through timely event detection.



SafeEdge alerts by building location with peak confidence

Key Recommendations

Residential fires continue to represent the highest category of fire incidents, underscoring the critical need for enhanced kitchen safety measures. Following a notable upward trend in fire occurrences post-2020, it is imperative that fire departments increase their vigilance and preventive strategies. Implementing edge-based AI detection systems, such as SafeEdge, can significantly improve response times by identifying fires over 30 seconds earlier than traditional smoke detectors, while multi-frame confirmation technology effectively minimizes false positives. Additionally, the integration of privacy-preserving face blur technology ensures compliance with data protection regulations, making these advanced systems both efficient and responsible.